

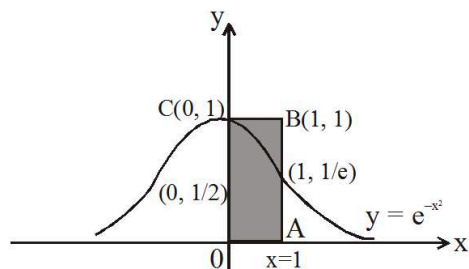
Exercise JA

Q1. Let S be the area of the region enclosed by e^{-x^2} and $x = 1$. Then :

(A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}} \right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}} \right)$

Ans. A, B, D

Sol. Area (OABC) = 1



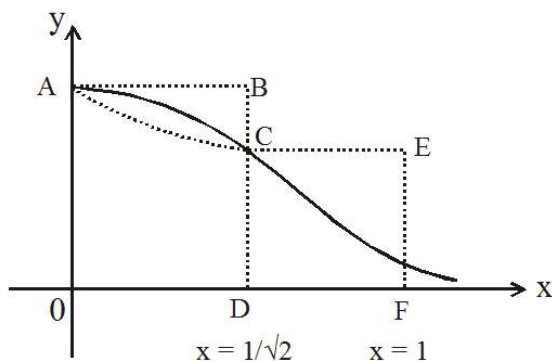
Shaded area is S.

Clearly $S < 1$

$$\text{and } \int_0^1 e^{-x^2} dx < \int_0^1 e^{-x} dx$$

$$\Rightarrow S > 1 - \frac{1}{e} \quad (\therefore \text{(B) is correct})$$

Again $S \geq \text{Area (trapezium ACDO)}$



$$\Rightarrow S \geq \frac{1}{2} \left(1 + \frac{1}{\sqrt{e}} \right) \left(\frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow S \geq \frac{1}{2\sqrt{2}} \left(1 + \frac{1}{\sqrt{e}} \right)$$

$\therefore$ C is wrong

Also $S \leq \text{Sum of areas of rectangles ABDO \& CEFD}$

$$\Rightarrow S \leq \frac{1}{\sqrt{2}} \times 1 + \left(1 - \frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{e}} \right)$$

$$\Rightarrow S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}}\right)$$

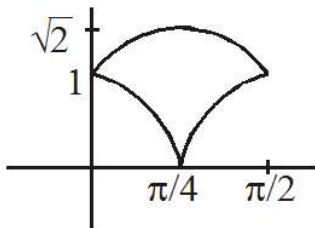
($\therefore$ (D) is correct)

Q2. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is :

(A) $4(\sqrt{2} - 1)$ (B) $2\sqrt{2}(\sqrt{2} - 1)$ (C) $2(\sqrt{2} + 1)$ (D) $2\sqrt{2}(\sqrt{2} + 1)$

Ans. B

Sol. $y = \sin x + \cos x = \sqrt{2} \sin \left(x + \frac{\pi}{4}\right)$



$$y = |\cos x - \sin x| = \sqrt{2} \left(\cos \left(x + \frac{\pi}{4}\right) \right)$$

Area

$$\begin{aligned} &= \int_0^{\pi/4} [(\sin x + \cos x) - (\cos x + \sin x)] dx \\ &= \int_{\pi/4}^{\pi/2} [(\sin x + \cos x) - (\sin x + \cos x)] dx \\ &= \left[-2\cos x \right]_{\pi/4}^{\pi/2} + \left[2\sin x \right]_{\pi/4}^{\pi/2} \\ &= 2\sqrt{2}(\sqrt{2} - 1) \end{aligned}$$

Q3. Let $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that

$f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then the value of $\int_{1/2}^1 f(x)$ lies in the interval :

(A) $(2e - 1, 2e)$ (B) $(2e - 1, 2e - 1)$ (C) $\left(\frac{e - 1}{2}, e - 1\right)$ (D) $\left(0, \frac{e - 1}{2}\right)$

Ans. D

Sol. $|f'(x) - 2f(x)| < 0$

Multiply both side by e^{-2x}

$$e^{-2x} |f'(x) - 2e^{-2x}|(x) < 0$$

$$\frac{d}{dx} (e^{-2x} |f'(x) - 2e^{-2x}|(x)) < 0$$

$$\text{Now, } g(x) = e^{-2x} |f'(x) - 2e^{-2x}|(x)$$

$\therefore g(x)$ is decreasing function.

$$x > \frac{1}{2}$$

$$g(x) < g\left(\frac{1}{2}\right)$$

$$\Rightarrow e^{-2x} |f'(x) - 2e^{-2x}|(x) < \frac{1}{e}$$

$$\Rightarrow |f'(x) - 2e^{-2x}|(x) < e^{2x-1}$$

$$\Rightarrow \int_{1/2}^1 |f'(x) - 2e^{-2x}|(x) dx > \frac{1}{e} \int_{1/2}^1 e^{2x} dx$$

$$= \left[\frac{1}{2e} e^{2x} \right]_{1/2}^1 = \frac{1}{2e} (e^2 - e) = \frac{1}{2} (e - 1)$$

$$\Rightarrow \int_{1/2}^1 |f'(x) - 2e^{-2x}|(x) dx < \frac{e - 1}{2}$$

$$\therefore \int_{1/2}^1 |f'(x) - 2e^{-2x}|(x) dx < 0$$

Q4.

Let $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$ for all $x \in \mathbb{R}$ and $f: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty)$ be a continuous function.

For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is :

Ans. (a) 3, (b) A - PRS , B - P , C - PQ, D - ST

Sol.

$$F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$$

$$F'(x) = 2 \left(\cos^2 \left(x^2 + \frac{\pi}{6} \right) \right) 2x - 2 \cos^2 x$$

$$\therefore F'(a) + 2 = 2 \int_0^a f(x) dx$$

$$2 \left(\cos \left(a^2 + \frac{\pi}{6} \right) \right)^2 2a - 2c \cos^2 a + 2 = \int_0^a f(x) dx$$

$$\Rightarrow 4 \cos^2 \left(a^2 + \frac{\pi}{6} \right) + 4a 2 \cos \left(a^2 + \frac{\pi}{6} \right) \cdot \left(-\sin \left(a^2 + \frac{\pi}{6} \right) \right) \times 2a + 4 \cos a \sin a = f(a)$$

$$\therefore f(0) = 4 \left(\frac{\sqrt{3}}{2} \right)^2 = 3$$

Q5. Match the following column :

Column I

Column II

(A) In a triangle ΔXYZ , let a , b and c be the lengths of the sides opposite to the

(P) 1

angles X , Y and Z , respectively. If $2(a^2 - b^2) = c^2$ and $\lambda = \frac{\sin(X - Y)}{\sin Z}$, then possible values of n for which $\cos(n\pi\lambda) = 0$ is (are)

(B) In a triangle ΔXYZ , let a , b and c be the lengths of the sides opposite to the angles

(Q) 2

X , Y and Z , respectively. If $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$, then possible value(s) of a/b is (are)

(C) In R^2 , let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1 - \beta)\hat{j}$ be the position vectors of X , Y and Z

(R) 3

with respect to the origin O , respectively. If the distance of Z from the bisector of the acute angle of $\overrightarrow{OX}$ with $\overrightarrow{OY}$ is $\frac{3}{\sqrt{2}}$, then possible value(s) of $|\beta|$ is (are)

(D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x = 0$, $x = 2$

(S) 5

$y^2 = 4x$ and $y = |\alpha x - 1| + |\alpha x - 2| + \alpha x$, where $\alpha \in \{0, 1\}$. Then the value(s) of

$F(\alpha) + \frac{8}{3}\sqrt{2}$, when $\alpha = 0$ and $\alpha = 1$, is (are)

Ans. A - PRS, B - P, C - PQ, D - ST

Sol. (A) $2(a^2 - b^2) = c^2$

$$2(4R^2(\sin^2 x - \sin^2 y)) = 4R^2 \sin^2 z$$

$$2(\sin(x - y) \sin(x + y)) = \sin^2 z$$

$$\frac{\sin(x - y)}{\sin z} = \frac{1}{2} = \lambda$$

$$\cos(n\pi\lambda) = \cos\left(\frac{n\pi}{2}\right) = 0$$

$n = \text{odd integer}$

$$(B) 1 + \cos 2x = 2 \cos 2y = 2 \sin x \sin y$$

$$\sin^2 x + \sin x \sin y - 2 \sin^2 y = 0$$

$$(\sin x - \sin y)(\sin x + 2 \sin y) = 0$$

$$\sin x = \sin y$$

$$a = b$$

$$\frac{a}{b} = 1$$

$$(C) \text{ distance of } z \text{ from bisector of } \overrightarrow{OX} \& \overrightarrow{OY} = \frac{3}{\sqrt{2}}$$

$$\left(\beta - \frac{1}{2}\right)^2 + \left(\beta - \frac{1}{2}\right)^2 = \frac{9}{2}$$

$$\beta = 2, -1$$

$$|\beta| = 2, 1$$

$$(D) \text{ when } \alpha = 0$$

$$A = 6 - \int_0^2 2\sqrt{x} dx$$

$$= 6 - \frac{8\sqrt{2}}{3}$$

$$\text{when } \alpha = 1$$

$$A = \int_0^1 (3 - x - 2\sqrt{x}) dx + \int_1^2 (x + 1 - 2\sqrt{x}) dx$$

$$= \left| 3x - \frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}} \right|_0^1 + \left| \frac{x^2}{2} + x - \frac{4}{3}x^{\frac{3}{2}} \right|_1^2$$

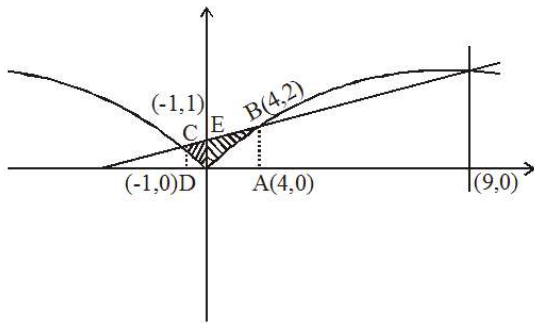
$$= 5 - \frac{8\sqrt{2}}{3}$$

Q6. Area of the region $\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\}$ is equal to

(A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

Ans. C

Sol. Shifting origin to $(-3, 0)$



$$\text{Area} \left\{ (x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x|}, 5y \leq x + 6 \leq 15 \right\}$$

Desired Area = Area of trapezium ABCD – [Area DCOD + Area OBAO]

$$\begin{aligned} &= \frac{1}{2} \times (1 + 2) \times 5 - \left(\int_{-1}^0 \sqrt{-x} dx + \int_0^4 \sqrt{x} dx \right) \\ &= \frac{15}{2} - \left(\frac{2}{3} + \frac{16}{3} \right) \\ &= \frac{3}{2} \end{aligned}$$

Q7. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in \mathbb{R}^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ into two equal parts, then:

(A) $0 < \alpha \leq \frac{1}{2}$ (B) $2\alpha^4 - 4\alpha^2 + 1 = 0$ (C) $\alpha^4 + 4\alpha^2 - 1 = 0$ (D) $\frac{1}{2} < \alpha < 1$

Ans. B, D

Sol. $y = x^3$

$$\int_0^1 (x - x^3) dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\int_0^\alpha (x - x^3) dx = \frac{1}{8}$$

$$4\alpha^2 - 2\alpha^4 = 1$$

$$2\alpha^4 - 4\alpha^2 + 1 = 0$$

$$2t^2 - 4t + 1 = 0$$

$$2t^2 - 4t + 1 = 0 \text{ (taking } t = \alpha^2 \text{)}$$

$$t = \frac{4 \pm \sqrt{16 - 8}}{4}$$

$$t = \frac{4 \pm 2\sqrt{2}}{4}$$

$$t = \alpha^2 = 1 \pm \frac{1}{\sqrt{2}} \quad \frac{1}{2} < \alpha < 1$$

Q8. (a) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that

$$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$$

for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ?

(A) The curve $y = f(x)$ passes through the point $(1, 2)$

(B) The curve $y = f(x)$ passes through the point $(2, -1)$

(C) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-2}{4}$

(D) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-1}{4}$

(b) A farmer F_1 has a land in the shape of a triangle with vertices at $P(0,0)$, $Q(1,1)$ and $R(2,0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ ($n > 1$). If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR , then the value of n is _____.

Ans. (a) BC, (b) 4

Sol.

$$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt \Rightarrow f(x) \cdot e^{-x} = (1 - 2x) \cdot e^{-x} + \int_0^x e^2 f(t) dt$$

$$\Rightarrow f(x) e^{-x} - e^{-x} \cdot f(x) = -2 \cdot e^{-x} - (1 - 2x) \cdot e^{-x} + e^{-x} \cdot f(x)$$

$$\Rightarrow f'(x) - 2f(x) = (2x - 3)$$

$$\text{I.F.} = e^{-2x}$$

$$\therefore y \cdot e^{-2x} = \int (2x - 3) \cdot e^{-2x} dx$$

$$\Rightarrow y \cdot e^{-2x} = (2x - 3) \cdot \frac{e^{-2x}}{-2} - 2 \int \frac{e^{-2x}}{-2} dx \Rightarrow y \cdot e^{-2x} = -\frac{(2x - 3)e^{-2x}}{2} - \frac{e^{-2x}}{2} + c$$

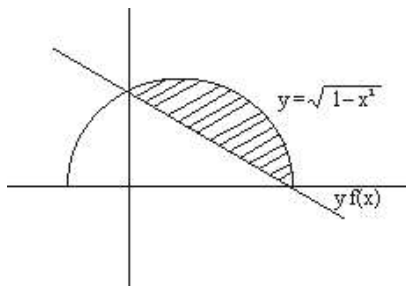
$$\Rightarrow y \cdot e^{-2x} = \frac{-(2x - 3) - 1}{2} + c \cdot e^{-2x} \Rightarrow y = (1 - x) + c \cdot e^{2x}$$

$$\Rightarrow y = (1 - x) + c \cdot e^{2x}$$

$$\text{put } x = 0$$

$$1 = 1 + c \Rightarrow c = 0$$

$$\therefore y = 1 - x \text{ which passes through point } (2, -1)$$



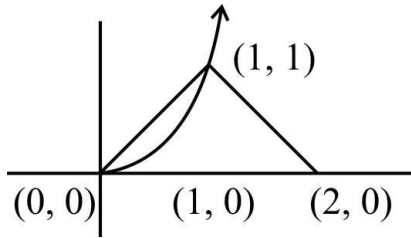
Now

$$\text{required area} = \frac{1}{4} \cdot \pi \cdot (1)^2 - \frac{1}{2} \cdot 1 \cdot 1 = \frac{\pi}{4} - \frac{1}{2}$$

- Q9. A farmer F_1 has a land in the shape of a triangle with vertices at $P(0, 0)$, $Q(1, 1)$ and $R(2, 0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$; ($n > 1$). If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR , then the value of n is

Ans. 4

Sol.



$$\text{Area} = \int_0^1 (x - x^n) dx = \frac{3}{10}$$

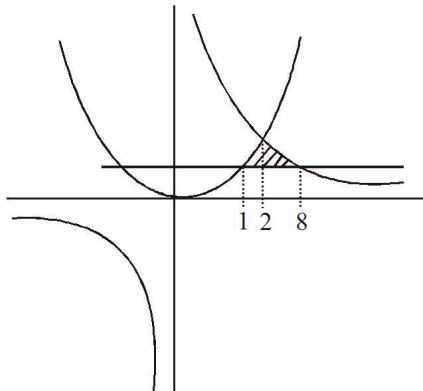
$$\left[\frac{x^2}{2} - \frac{x^{n+1}}{n+1} \right]_0^1 = \frac{3}{10} \cdot \frac{1}{2} - \frac{1}{n+1} = \frac{3}{10} \therefore n+1 = 5 \Rightarrow n = 4$$

- Q10. The area of the region $\{(x,y): xy \leq 8, 1 \leq y \leq x^2\}$ is

(A) $16\log_e 2 - 6$ (B) $8\log_e 2 - \frac{14}{3}$ (C) $16\log_e 2 - \frac{14}{3}$ (D) $8\log_e 2 - \frac{7}{3}$

Ans. C

Sol.



$$x^3 = 8$$

$$x = 2$$

$$\begin{aligned} \text{Area} &= \int_1^2 (x^2 - 1) dx + \int_2^8 \left(\frac{8}{x} - 1 \right) dx \\ &= \left(\frac{x^3}{3} - x \right)_1^2 + \left(8 \ln x - x \right)_2^8 \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{8}{3} - 2 - \frac{1}{3} + 1 \right) + [8 \ln 8 - 8 \ln 2 - 8 + 2] \\
&= \left(\frac{7}{3} - 1 \right) + 8 \ln 4 - 6 \\
&= 16 \ln 2 + \frac{7}{3} - 7 \\
&= 16 \ln 2 - \frac{14}{3}
\end{aligned}$$

Q11. Let the functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$.

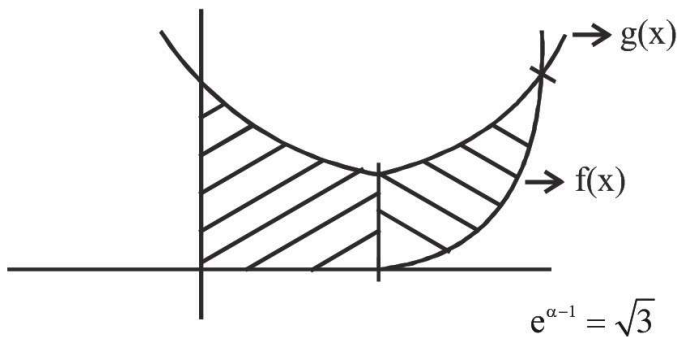
Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is

- (A) $(2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$ (B) $(2 + \sqrt{3}) + \frac{1}{2}(e - e^{-1})$ (C) $(2 - \sqrt{3}) + \frac{1}{2}(e + e^{-1})$
 (D) $(2 + \sqrt{3}) + \frac{1}{2}(e + e^{-1})$

Ans. A

Sol. $f(x) \begin{cases} \rightarrow e^{x-1} - e^{1-x} & x \geq 1 \\ \rightarrow 0 & x < 1 \end{cases}$

$$g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$$



$$P\left(1 + \ln \sqrt{3}, \frac{2}{\sqrt{3}}\right) \equiv P\left(\alpha, \frac{2}{\sqrt{3}}\right)$$

$$\text{Required Area} = \int_0^1 \frac{e^{x-1} + e^{1-x}}{2} dx + \int_1^\alpha f(x) - f(x) dx$$

$$= \frac{1}{2} \left(e - \frac{1}{e} \right) + \int_1^\alpha \left(\frac{e^{x-1} + e^{1-x}}{2} - (e^{x-1} - e^{1-x}) \right) dx$$

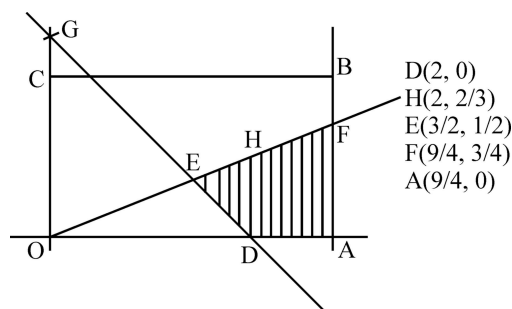
$$= \frac{1}{2} \left(e - \frac{1}{e} \right) + \int_1^\alpha \left(\frac{3}{2} e^{1-x} - \frac{e^{x-1}}{2} \right) dx = (2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$$

Q12. The area of the region $\left\{ (x, y) : 0 \leq x \leq \frac{9}{4}, \quad 0 \leq y \leq 1, \quad x \geq 3y, \quad x + y \geq 2 \right\}$ is

- (A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

Ans. A

Sol.



$$AB \equiv x = \frac{9}{4}$$

$$BC \equiv y = 1$$

$$DG \equiv x + y = 2$$

$$OF \equiv x = 3y$$

$$DH \equiv x = 2$$

we will divide shaded region in two parts

required area = Area of $\triangle DEH$ + area of trapezium AFHD

$$= \frac{1}{2} \left(\frac{2}{3} \right) \left(\frac{1}{2} \right) + \frac{1}{2} \left(\frac{2}{3} + \frac{3}{4} \right) \left(\frac{1}{4} \right) = \frac{11}{32}$$

Q13. Consider the functions $f, g : R \rightarrow R$ defined by $f(x) = x^2 + \frac{5}{12}$ and

$$g(x) = \begin{cases} 2 \left(1 - \frac{4|x|}{3} \right) & |x| \leq \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$$

If α is the area of the region $\left\{ (x, y) \in R \times R : |x| \leq \frac{3}{4}, 0 \leq y \leq \min\{f(x), g(x)\} \right\}$ then the value of 9α is ____.

Ans. 6

Sol.

$$f(x) = x^2 + \frac{5}{12}$$

$$g(x) = 2 \left(1 - \frac{4}{3}|x| \right)$$

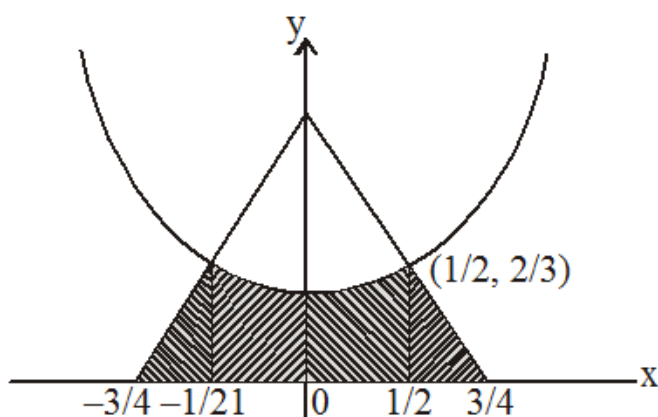
$$|x| \leq \frac{3}{4} \text{ and } 0 \leq y \leq (f(x), g(x))$$

for $x \geq 0$

$$x^2 + \frac{5}{12} = 2 - \frac{8}{3}x$$

$$\Rightarrow 12x^2 + 32x - 19 = 0$$

$$12x^2 + 38x - 6x - 19 = 0 \Rightarrow x = \frac{1}{2}$$



$$\text{Required area} = 2 \left(\int_0^{1/2} \left(x^2 + \frac{5}{12} \right) dx + \frac{1}{2} \times \frac{1}{4} \times \left(2 - \frac{4}{3} \right) \right)$$

$$\Rightarrow \alpha = 2 \left(\frac{1}{24} + \frac{5}{24} + \frac{2}{24} \right) = \frac{2}{3} \text{ sq. units} \Rightarrow 9\alpha = 9 \times \frac{2}{3} = 6$$

Q14. Let $f : [0, 1] \rightarrow [0, 1]$ be the function defined by $f(x) = \frac{x^3}{3} - x^2 + \frac{5}{9}x + \frac{17}{36}$. Consider the square region $S = [0, 1] \times [0, 1]$. Let $G = \{(x, y) \in S : y > f(x)\}$ be called the green region and $R = \{(x, y) \in S : y < f(x)\}$ be called the red region. Let $L_h = \{(x, h) \in S : x \in [0, 1]\}$ be the horizontal line drawn at a height $h \in [0, 1]$. Then which of the following statements is(are) true?

- (A) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3} \right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h
- (B) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3} \right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h
- (C) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3} \right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h
- (D) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3} \right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h

Ans. B, C, D

Sol. $f(x) = \frac{x^3}{3} - x^2 + \frac{5x}{9} + \frac{17}{36}$

$$f'(x) = x^2 - 2x + \frac{5}{9}$$

$$f'(x) = 0 \text{ at } x = \frac{1}{3} \text{ in } [0, 1]$$

A_R = Area of Red region

A_G = Area of Green region

$$A_R = \int_0^1 f(x)dx = \frac{1}{2}$$

Total area = 1

$$\Rightarrow A_G = \frac{1}{2}$$

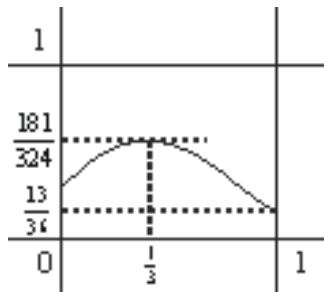
$$\int_0^1 f(x)dx = \frac{1}{2}$$

$$A_G = A_R$$

$$f(0) = \frac{17}{36}$$

$$f(1) = \frac{13}{36} \approx 0.36$$

$$\left(\frac{1}{3}\right) = \frac{181}{324} \approx 0.558$$



(A) Given statement would be correct when $h = \frac{3}{4}$ but $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$

$\Rightarrow$ (A) is incorrect

(B) Given statement would be correct when $h = \frac{1}{4}$

$\Rightarrow$ (B) is correct

(C) When $h = \frac{181}{324}$, $A_R = \frac{1}{2}$, $A_G < \frac{1}{2}$

$$h = \frac{13}{36}, A_R < \frac{1}{2}, A_G = \frac{1}{2}$$

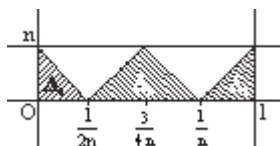
$\Rightarrow A_R = A_G$ for some $h \in \left(\frac{13}{36}, \frac{181}{324}\right)$

$\Rightarrow$ (C) is correct

(D) Option (D) is remaining coloured part of option (C), hence option (D) is also correct.

Q15. Let $n \geq 2$ be a natural number and $f: [0, 1] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} n(1 - 2nx) & \text{if } 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx - 1) & \text{if } \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1 - nx) & \text{if } \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx - 1) & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$



If n is such that the area of the region bounded by the curves $x = 0$, $x = 1$, $y = 0$ and $y = f(x)$ is 4, then the maximum value of the function f is _____.

Ans. 8

Sol.

$$f(x) = \begin{cases} n(1 - 2nx) & 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx - 1) & \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1 - nx) & \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx - 1) & \frac{1}{n} \leq x \leq 1 \end{cases}$$

$$\Delta_1 + \Delta_2 + \Delta_3 = 4$$

$$\frac{1}{2}n \times \frac{1}{2n} + \frac{1}{2} \times n \times \frac{1}{2n} + \frac{1}{2} \times n \times \left(1 - \frac{1}{n}\right) = 4$$

$$4 = \frac{n}{2}$$

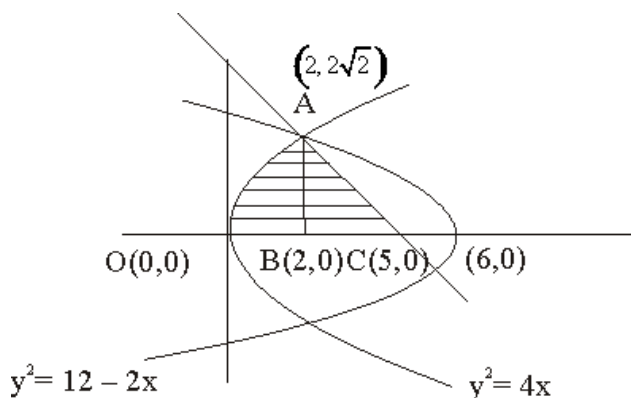
$$n = 8$$

Q16. Let $S = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0, y \geq 0, y^2 \leq 4x, y^2 \leq 12 - 2x \text{ and } 3y + \sqrt{8}x \leq 5\sqrt{8}\}$ If the area of the region S is $\alpha\sqrt{2}$, then α is equal to :

(A) $\frac{17}{2}$ (B) $\frac{17}{3}$ (C) $\frac{17}{4}$ (D) $\frac{17}{5}$

Ans. B

Sol. $3y + \sqrt{8}x = 5\sqrt{8}$



$$4x = 12 - 2x \Rightarrow 6x = 12 \Rightarrow x = 2$$

$$x = 2 \Rightarrow y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$$

line $3y + \sqrt{8}x = 5\sqrt{8}$ passes through $A(2, 2\sqrt{2})$ & $C(5,0)$

Required area = Area of OAB + Area of ΔABC

$$\begin{aligned}
&= \int_0^2 2\sqrt{x} dx + \frac{1}{2}(5-2)(2\sqrt{2}) \\
&= \frac{4}{3} \left(x^{\frac{3}{2}} \right)_0^2 + 3\sqrt{2} \\
&= \frac{8}{3}\sqrt{2} + 3\sqrt{2} \\
&= \frac{17}{3}\sqrt{2}
\end{aligned}$$

Exercise C1

Q1. The area bounded by regions $y = \sqrt{x}$ & $x = -\sqrt{y}$ and the curve $x^2 + y^2 = 2$ is (above x-axis) :

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2} - \frac{1}{3}$ (C) $\frac{\pi}{2}$ (D) None

Ans. C

Sol. -

Q2. Area bounded by the curve $y = \max\{\sin x, \cos x\}$ and x-axis, between the lines $x = \frac{\pi}{4}$ and $x = 2\pi$ is equal to

- (A) $\frac{(4\sqrt{2} - 1)}{\sqrt{2}}$ sq. units (B) $(4\sqrt{2} - 1)$ sq. units (C) $\frac{(4\sqrt{2} - 1)}{2}$ sq. units

(D) None of these

Ans. C

Sol. -

Q3. If $A(n)$ represents the area bounded by the curve $y = n \cdot \ln x$, where $n \in N$ and $n > 1$, the x-axis and the lines $x = 1$ and $x = e$, then the value of $A(n) + nA(n-1)$ is equal to

- (A) $\frac{n^2}{e+1}$ (B) $\frac{n^2}{e-1}$ (C) n^2 (D) en^2

Ans. C

Sol. -

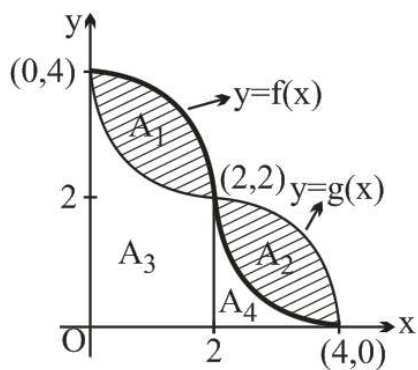
Q4. Suppose $y = f(x)$ and $y = g(x)$ are two continuous functions whose graphs intersect at the three points $(0, 4)$, $(2, 2)$ and $(4, 0)$ with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$.

If $\int_0^4 [f(x) - g(x)] dx = 10$ and $\int_2^4 [g(x) - f(x)] dx = 5$, the area between two curves for $0 < x < 2$, is

- (A) 5 (B) 10 (C) 15 (D) 20

Ans. C

Sol.



$$\text{Given } \int_0^4 f(x) dx - \int_0^4 g(x) dx = 10$$

$$(A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$$

$$\text{again } \int_2^4 g(x) dx - \int_0^2 f(x) dx = 5$$

$$(A_2 + A_4) - A_4 = 5$$

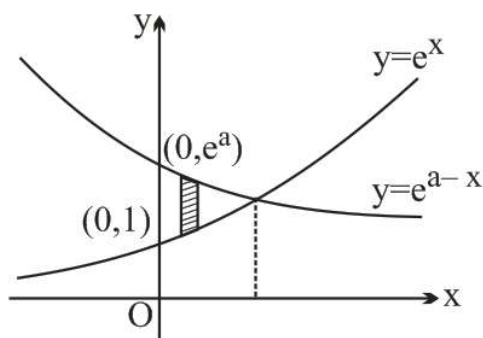
$$A_2 = 5 \quad \dots\dots\dots(2)$$

$$\therefore A_1 = 15 \text{ Ans.}$$

- Q5. Let 'a' be a positive constant number. Consider two curves $C_1: y = e^x$, $C_2: y = e^{a-x}$. Let S be the area of the part surrounding by C_1 , C_2 and the y-axis, then $\lim_{a \rightarrow 0} \frac{S}{a^2}$ equals
- (A) 4 (B) 1/2 (C) 0 (D) 1/4

Ans. D

Sol.



Solving $e^x = e^{a-x}$, we get

$$e^{2x} = e^a \quad \Rightarrow \quad x = \frac{a}{2}$$

$$S = \int_0^{a/2} (e^a \cdot e^{-x} - e^x) dx = [- (e^a \cdot e^{-x} + e^x)]_0^{a/2}$$

$$= (e^a + 1) - (e^{a/2} + e^{a/2}) = e^a - 2e^{a/2} + 1 = (e^{a/2} - 1)^2$$

$$\therefore \frac{S}{a^2} = \left(\frac{e^{a/2} - 1}{a} \right)^2 = \frac{1}{4} \left(\frac{e^{a/2} - 1}{a/2} \right)^2$$

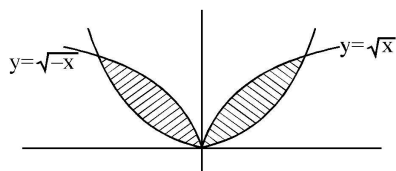
$$\therefore \lim_{a \rightarrow \infty} \frac{S}{a^2} = \frac{1}{4} \text{ Ans.}$$

Q6. The area of the region(s) enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is

- (A) $1/3$ (B) $2/3$ (C) $1/6$ (D) 1

Ans. B

Sol.



$$A = \left(\frac{16ab}{3} \right) \cdot 2$$

$$a = \frac{1}{4}; b = \frac{1}{4}$$

$$A = \frac{2}{3} \text{ Ans.}$$

Q7. The slope of the tangent to a curve $y = f(x)$ at $(x, f(x))$ is $2x + 1$. If the curve passes through the point $(1, 2)$ then the area of the region bounded by the curve, the x-axis and the line $x = 1$ is

- (A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{1}{6}$ (D) 1

Ans. A

Sol. -

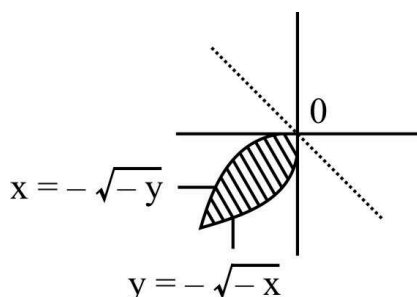
Q8. The area bounded by the curves $y = -\sqrt{-x}$ and $x = -\sqrt{-y}$ where $x, y \leq 0$

- (A) cannot be determined (B) is $1/3$ (C) is $2/3$

(D) is same as that of the figure bounded by the curves $y = \sqrt{-x}$; $x \leq 0$ and $x = \sqrt{-y}$; $y \leq 0$

Ans. B

Sol. Given that, $y = -\sqrt{-x}$



$\Rightarrow y^2 = -x$, where x and y both are negative.

Now, $x = -\sqrt{-y}$

$\Rightarrow x^2 = -y$, where x and y both are negative.

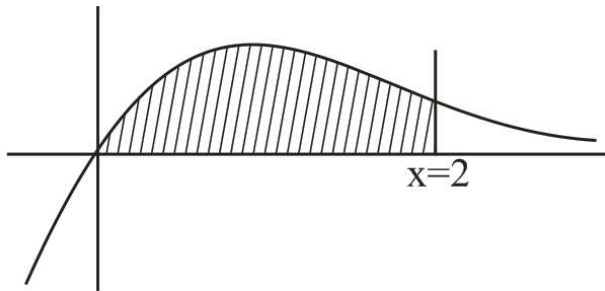
$$\therefore \text{Area} = \left| \int_{-1}^0 -\sqrt{-x} \, dx - \int_{-1}^0 -x^2 \, dx \right| = \frac{1}{3}$$

Q9. The area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x -coordinate of the curve's inflection point, is

- (A) $1 - 3e^{-2}$ (B) $1 - 2e^{-2}$ (C) $1 - e^{-2}$ (D) 1

Ans. A

Sol.



$$y = x e^{-x}$$

$$y' = e^{-x} - x e^{-x} = (1 - x)e^{-x} \uparrow \text{for } x < 1$$

$$y'' = -e^{-x} [e^{-x} - x e^{-x}] = e^{-x} [-1 - 1 + x]$$

$$= (x - 2)e^{-x}$$

$$\text{for point of inflection } y'' = 0 \quad \Rightarrow \quad x = 2$$

$$A = \int_0^2 x e^{-x} \, dx = x e^{-x} \Big|_0^2 - \int_0^2 e^{-x} \, dx$$

$$= (-2 e^{-2}) - \left(e^{-x} \right)_0^2$$

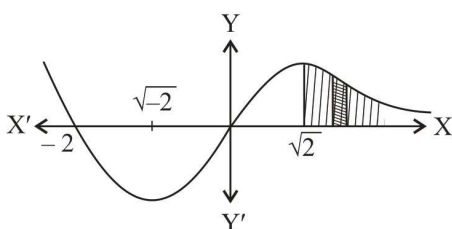
$$= -2 e^{-2} - (e^{-2} - 1) = -e^{-2} - 2e^{-2} = 1 - 3e^{-2} \text{ Ans.}$$

Q10. Area enclosed by the curve $y = (x^2 + 2x)e^{-x}$ and the positive x -axis is

- (A) 1 (B) 2 (C) 4 (D) 6

Ans. C

Sol.



Required area $A = \int_0^{\infty} e^{-x}(x^2 + 2x)dx = 4$ Ans.

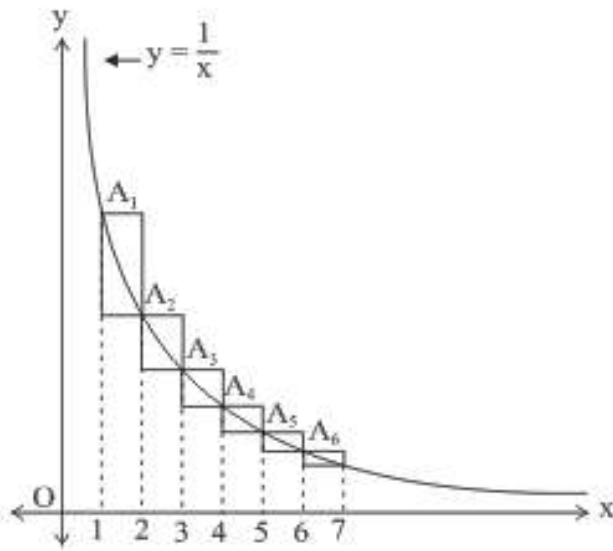
- Q11. Infinite rectangles each of width 1 unit and height $\left(\frac{1}{n} - \frac{1}{n+1}\right)$ ($n \in \mathbb{N}$) are constructed such that ends of exactly one diagonal of every rectangle lies along the curve $y = \frac{1}{x}$. The sum of areas of all such rectangles, is
- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{2}{4}$ (D) 1

Ans. D

Sol. **Method -I:**

$$\text{Required sum} = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \right] \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1$$

Method - II :



$$\text{Area of } r^{\text{th}} \text{ rectangle } A_r = (\text{height}) \times (\text{width}) = \left(\frac{1}{r} - \frac{1}{r+1}\right) \cdot 1$$

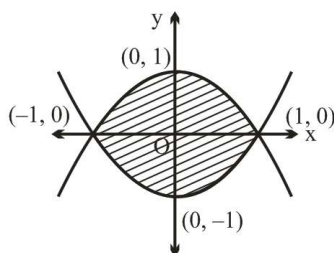
$$\begin{aligned} \text{Sum of area} &= \lim_{n \rightarrow \infty} \sum_{r=1}^n A_r = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \right] \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1 \end{aligned}$$

- Q12. Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$, is

- (A) 3 (B) $\frac{8}{3}$ (C) $\frac{7}{3}$ (D) $\frac{10}{3}$

Ans. B

Sol.



We have $y^2 = (x^2 - 1)^2 \Rightarrow y = x^2 - 1 \text{ or } y = 1 - x^2$

So, required area $= 4 \int_0^1 (1 - x^2) dx = 4 \left[x - \frac{x^3}{3} \right]_0^1 = \frac{8}{3}$

Paragraph

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$.

\n

Q13. Let $f: [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1 - x)$ for all $x \in [-1, 2]$.

Let $R_1 = \int_{-1}^2 xf(x)dx$, and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$, and the x -axis. Then :

- (A) $R_1 = 2R_2$ (B) $R_1 = 3R_2$ (C) $2R_1 = R_2$ (D) $3R_1 = R_2$

Ans. C

Sol.

$$R_2 = \int_{-1}^2 f(x) dx, R_1 = \int_{-1}^2 xf(x) dx$$

$$\int_{-1}^2 (1-x)f(1-x) dx \left(\because \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right)$$

$$\int_{-1}^2 (1-x)f(x) dx \quad (\text{given } f(x) = f(1-x))$$

$$= \int_{-1}^2 f(x) dx - \int_{-1}^2 xf(x) dx$$

or $R_1 = R_2 - R_1 \Rightarrow 2R_1 = R_2$